

- 3** Oil seed rape (canola), *Brassica napus*, has been genetically modified to be resistant to herbicides containing glufosinate ammonium. The genetically modified (GM) oil seed rape contains the *bar* gene, obtained from a soil bacterium. This gene codes for an enzyme that converts glufosinate ammonium into a non-toxic compound.

**(a)** Outline the advantages to farmers of growing glufosinate-resistant oil seed rape.

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**(b)** The *bar* gene was introduced into the oil seed rape using plasmids. The plasmids also contained a promoter taken from thale cress, *Arabidopsis thaliana*.

**(i)** Outline the structure of a plasmid.

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**(ii)** Explain how the properties of plasmids make them suitable for use during genetic modification programmes.

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(iii) Describe the role of a promoter in gene expression.

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- (c) The pollen of oil seed rape is transferred from one flower to another by insects. After pollination, fertilisation and seed formation can occur. One of the potential problems of growing glufosinate-resistant oil seed rape is that pollen from these plants could be transferred to the flowers of wild relatives, such as wild radish, *Raphanus raphanistrum*. This could result in genetic changes in these wild species.

An experiment was carried out to investigate whether glufosinate-resistant hybrids between GM oil seed rape and wild radish plants are likely to compete successfully with non-hybrid or non-resistant plants in the natural environment.

- Type 1 hybrids were produced by transferring pollen from wild radish (diploid number 18) to glufosinate-resistant oil seed rape (diploid number 38).
- Type 2 hybrids were produced by transferring pollen from glufosinate-resistant oil seed rape to wild radish.
- Each hybrid was then crossed with wild radish over several generations.
- The resulting offspring were then grown in field trials, together with normal wild radish.
- The height of the plants and number of seeds each produced were measured. Then the plants were tested for the *bar* gene.

Table 3.1 shows the results.

**Table 3.1**

| type of plant                                | number of seeds per plant | mean height /cm | presence of <i>bar</i> gene |
|----------------------------------------------|---------------------------|-----------------|-----------------------------|
| offspring from type 1 hybrid and wild radish | 265                       | 22.3            | absent                      |
|                                              | 99                        | 28.3            | present                     |
| offspring from type 2 hybrid and wild radish | 3958                      | 88.7            | absent                      |
|                                              | 2047                      | 95.0            | present                     |
| wild radishes                                | 3515                      | 76.5            | absent                      |

- (i) Predict the diploid number of chromosomes in a hybrid between oil seed rape and wild radish.

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- (ii) Suggest how the researchers could have determined whether or not the *bar* gene was present in the plants.

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- (iii) Many varieties of GM oil seed rape are male sterile, meaning that they do not produce pollen.

With reference to Table 3.1, suggest the advantages to the environment of growing male sterile varieties of GM oil seed rape, rather than GM varieties that produce pollen.

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